

Week 1: Bulgarian solitaire

Facilitator guide for the three adults (W1-F-v1)

Contents: 1 materials and preparation; 2 the whole-group launch; 3 the K–1 table; 4 the grades 2–3 table; 5 the grades 4–5 table; 6 the share and closing; 7 the mathematics behind the session. Sections 3–5 hold the order of problems per table with a rough clock, complete answers and arguments for every problem, hints in the order to give them, and what to do when a table stalls or finishes early.

The one move used all hour: counters sit in piles; take one counter from every pile and put all the counters you took together as one new pile; a pile that becomes empty is gone. An arrangement is written biggest pile first, with hyphens: 4-2, 3-2-1, 1-1-1-1-1-1. The K–1 pages use pictures and words only; the adult at that table writes tower sizes on the blanks.

1. Materials and preparation

Materials, with counts:

- Counters, one color, about 150: ten per K–1 child (30), fifteen per child at the other tables (60 for grades 2–3, 45 for grades 4–5), six for the launch, the rest spare. Flat counters that stack are best.
- One deck of playing cards; only the ace to 10 of one suit are used (grades 4–5, Problem 13).
- Crayons or thick pencils, one per K–1 child; pencils and two erasers per other table.
- Printed pages: the K–1 set (plate mat plus problem pages), one per K–1 child, and one spare; the grades 2–3 set, one per child (four); the grades 4–5 set, one per child (three); this guide, one per adult.
- For the launch: a table or floor space all ten children can see, and a whiteboard or large sheet to write four arrangements on.

Preparation (ten minutes before): put a plate mat, ten counters and a crayon at each K–1 seat; fifteen counters and a pencil at each other seat; the page sets face down. Suggested adults: the parent at the K–1 table (its guide below is a full script; the problems need reading aloud and writing, not mathematics on the spot), the organizer at grades 2–3 (four children, the busiest table), the second mathematician at grades 4–5 (the table with the two written arguments). Each adult reads the launch and the first four problems of their table before the session and keeps the answer tables in this guide open.

Session shape: 0:00–0:05 free handling (children make piles, stack, sort; the K–1 adult shows the plate mat and lets them build any piles on it); 0:05–0:08 the launch; 0:08–0:50 table work; 0:50–0:58 the share; 0:58–1:00 pack up.

2. The whole-group launch (two minutes)

Everyone stands round one table. The organizer has six counters in three piles of two.

Say and do: “Here is the one move we will use all hour. Take one counter from every pile” (take them, saying “one, two, three”), “and put all the counters you took together as a new pile.” Now there are piles of 3, 1, 1, 1. “Again: one from every pile. Four piles, so four counters. New pile.” Now 4 and 2. “Again.” Now

3, 2, 1. “Again.” Take one from each of the three piles, make a pile of three: it is 3, 2, 1 again. “Nothing changed.”

Write on the board, saying “we write the piles biggest first”: $2-2-2 \rightarrow 3-1-1-1 \rightarrow 4-2 \rightarrow 3-2-1 \rightarrow 3-2-1$.

Say: “It stopped changing. Does that happen from every start? With every number of counters? How long does it take? Each table is going to find out. Off you go.” Do not say the word staircase, do not mention triangular numbers, and do not say what other numbers of counters do.

For the K–1 children, their adult adds at the table: “We will put our piles on plates, and after every move we will color a picture of the towers.”

Verified: 2-2-2 has 3 piles, so the new pile is 3; 3-1-1-1 has 4 piles, new pile 4; 4-2 has 2 piles, new pile 2; 3-2-1 has 3 piles, new pile 3, and the picture is unchanged.

3. K–1 table: script, answers, hints

Order: Problems 1–9 are the core (about 40 minutes); then 10, 11, 12 in that order; 13–15 for a fast group. Rough clock, with table work starting at 0:08: Problems 1–2 done by 0:16, 3–4 by 0:25, 5–7 by 0:40, 8–9 by 0:48; a group that is ahead goes on to 10, 11, 12. Each child has their own pages, so a faster child can be read the next problem while the others finish. Every problem: read the sentence, wait for the action, then read the next sentence. Write tower sizes under each frame as the child finishes it (for example “2-1”), and write the numbers the child says on the blanks. If coloring gets slow, color the frames yourself while the child says the tower heights, or have the child color only the last picture of a run while you write the sizes under the others. A child may color the towers in any order; write the sizes sorted, biggest first, so pictures can be compared.

The move on plates: “one counter off every plate that has counters, all together on an empty plate.” Common slip: taking counters from only some plates, or putting the new pile on a plate that still has counters. Fix: have the child point to each plate with counters before taking, and say “empty plate” out loud.

The point of the whole table, for you: three counters always end at 2-1; four counters never stop and go round three pictures; six counters always end at 3-2-1, from every one of the eleven starts, in zero to six moves; the arrangements that stay still are the staircases 2-1, 3-2-1, 4-3-2-1 with 3, 6, 10 counters; the new pile is always as big as the number of plates that had counters, because one counter came off each plate.

Complete tables you will need.

Three counters, three ways: $3 \rightarrow 2-1$; $2-1 \rightarrow 2-1$ (stays); $1-1-1 \rightarrow 3$. Every start ends at 2-1; 1-1-1 takes 2 moves, 3 takes 1, 2-1 takes 0. Nothing becomes 1-1-1.

Four counters, five ways: $4 \rightarrow 3-1$; $3-1 \rightarrow 2-2$; $2-2 \rightarrow 2-1-1$; $2-1-1 \rightarrow 3-1$; $1-1-1-1 \rightarrow 4$. Nothing stays still. Every start goes round the three pictures 3-1, 2-2, 2-1-1 for ever. Nothing becomes 1-1-1-1; 3-1 comes from two starts (4 and 2-1-1).

Five counters, seven ways: $5 \rightarrow 4-1 \rightarrow 3-2 \rightarrow 2-2-1 \rightarrow 3-1-1 \rightarrow 3-2 \rightarrow \dots$, a loop of three pictures (3-2, 2-2-1, 3-1-1) entered after 2 moves; $1-1-1-1-1 \rightarrow 5$ (then as above); $2-1-1-1 \rightarrow 4-1$. Nothing stays still.

Six counters, eleven ways, all end at 3-2-1. Moves to get there: 3-2-1: 0; 4-2: 1; 5-1: 2; 3-1-1-1: 2; 6: 3; 2-2-2: 3; 2-1-1-1-1: 3; 1-1-1-1-1-1: 4; 3-3: 4; 4-1-1: 5; 2-2-1-1: 6. The runs: $6 \rightarrow 5-1 \rightarrow 4-2 \rightarrow 3-2-1$. $1-1-1-1-1-1 \rightarrow 6 \rightarrow 5-1 \rightarrow 4-2 \rightarrow 3-2-1$. $3-3 \rightarrow 2-2-2 \rightarrow 3-1-1-1 \rightarrow 4-2 \rightarrow 3-2-1$. $2-2-1-1 \rightarrow 4-1-1 \rightarrow 3-3 \rightarrow 2-2-2 \rightarrow 3-1-1-1 \rightarrow 4-2 \rightarrow 3-2-1$. $2-1-1-1-1 \rightarrow 5-1 \rightarrow 4-2 \rightarrow 3-2-1$. Every run passes through 4-2 just before 3-2-1. The same facts as a table:

start	moves	run
3-2-1	0	stays 3-2-1
4-2	1	4-2 $\rightarrow$ 3-2-1
5-1	2	5-1 $\rightarrow$ 4-2 $\rightarrow$ 3-2-1
3-1-1-1	2	3-1-1-1 $\rightarrow$ 4-2 $\rightarrow$ 3-2-1
6	3	6 $\rightarrow$ 5-1 $\rightarrow$ 4-2 $\rightarrow$ 3-2-1
2-2-2	3	2-2-2 $\rightarrow$ 3-1-1-1 $\rightarrow$ 4-2 $\rightarrow$ 3-2-1
2-1-1-1-1	3	2-1-1-1-1 $\rightarrow$ 5-1 $\rightarrow$ 4-2 $\rightarrow$ 3-2-1
1-1-1-1-1-1	4	1-1-1-1-1-1 $\rightarrow$ 6 $\rightarrow$ 5-1 $\rightarrow$ 4-2 $\rightarrow$ 3-2-1
3-3	4	3-3 $\rightarrow$ 2-2-2 $\rightarrow$ 3-1-1-1 $\rightarrow$ 4-2 $\rightarrow$ 3-2-1
4-1-1	5	4-1-1 $\rightarrow$ 3-3 $\rightarrow$ 2-2-2 $\rightarrow$ 3-1-1-1 $\rightarrow$ 4-2 $\rightarrow$ 3-2-1
2-2-1-1	6	2-2-1-1 $\rightarrow$ 4-1-1 $\rightarrow$ 3-3 $\rightarrow$ 2-2-2 $\rightarrow$ 3-1-1-1 $\rightarrow$ 4-2 $\rightarrow$ 3-2-1

Ten counters on one plate: 10 $\rightarrow$ 9-1 $\rightarrow$ 8-2 $\rightarrow$ 7-2-1 $\rightarrow$ 6-3-1 $\rightarrow$ 5-3-2 $\rightarrow$ 4-3-2-1, six moves, and 4-3-2-1 stays still.

Problem 1 (1-1-1). Expect 1-1-1 $\rightarrow$ 3 $\rightarrow$ 2-1 $\rightarrow$ 2-1: pictures “three towers of one”, “one tower of three”, “towers 2 and 1”, “towers 2 and 1”. The count is three every time (nothing is lost; counters only move). The first picture that looks just like the one after it is the picture after 2 moves. Ask: “Will it ever change again? Try once more.” (It stays 2-1.) Hint if the move goes wrong: “Every plate with counters gives one counter. Point to each one.”

Problem 2 (all starts with three). 3 $\rightarrow$ 2-1 $\rightarrow$ 2-1 (stopped after 1 move); 2-1 $\rightarrow$ 2-1 (stopped before any move). The only other way is one counter on each of three plates, which was Problem 1, so there is no new way: the three ways are 3, 2-1 and 1-1-1, and every one of them ends at 2-1. Say: “We tried every way to put down three counters. Every way ends with towers 2 and 1.” This is the whole proof for three counters. If a child proposes “one and two” as new, put 2-1 and 1-2 side by side on plates and ask whether the towers look different.

Problem 3 (four on one plate). 4 $\rightarrow$ 3-1 $\rightarrow$ 2-2 $\rightarrow$ 2-1-1 $\rightarrow$ 3-1 $\rightarrow$ 2-2 $\rightarrow$ 2-1-1. Same pictures: after 1 and after 4 moves (3-1); after 2 and after 5 (2-2); after 3 and after 6 (2-1-1). It never stops changing; it goes round three pictures for ever. Ask: “If we made a hundred moves, what picture would we see? One of these three.” Hint for finding pairs: “Look at the tallest tower in each picture.”

Problem 4 (one on each of four plates). 1-1-1-1 $\rightarrow$ 4 $\rightarrow$ 3-1 $\rightarrow$ 2-2 $\rightarrow$ 2-1-1. After 1 move it is the start of Problem 3 (four on one plate), so from then on it is the same run, round the same three pictures. Say: “Two different starts, and after one move they are the same. From then on they can never be different again, because the move always does the same thing to the same picture.”

Problem 5 (six on one plate). 6 $\rightarrow$ 5-1 $\rightarrow$ 4-2 $\rightarrow$ 3-2-1, then 3-2-1 again: three moves; towers 3, 2 and 1. Ask: “Four counters never stopped. Six counters did. Where?”

Problem 6 (one on each of six plates). After one move: six on one plate, which is how Problem 5 started. It stops after 4 moves at 3-2-1, the same last picture; one move more than Problem 5, because the first move made Problem 5’s start.

Problem 7 (3-3, then 2-2-2). 3-3 $\rightarrow$ 2-2-2 $\rightarrow$ 3-1-1-1 $\rightarrow$ 4-2 $\rightarrow$ 3-2-1: four moves. 2-2-2 $\rightarrow$ 3-1-1-1 $\rightarrow$ 4-2 $\rightarrow$ 3-2-1: three moves (this is the run from the launch). The picture after 1 move in the first run is the start of the second run, so the second run is the first run with its first move cut off: one move fewer. Ask: “So

far every start with six counters has ended at 3, 2, 1. Do you think every start does?" Do not answer yet; Problem 12 and the table above answer it.

Problem 8 (guess the new pile). 2-2-2: 3 plates with counters, new pile 3 (giving 3-1-1-1); 4 plates, new pile 4 (4-2); 2 plates, new pile 2 (3-2-1); 3 plates, new pile 3 (3-2-1, unchanged). Write the pairs 3/3, 4/4, 2/2, 3/3. 3-3: 2 plates, new pile 2 (2-2-2); then 3/3 (3-1-1-1); 4/4 (4-2); 2/2 (3-2-1). The new pile always has exactly as many counters as there were plates with counters. Ask: "Why is it always the same number?" Expected: "because I took one from each plate." That sentence is the law of the game; you will use it in Problems 9, 10 and 15.

Problem 9 (which arrangement stays still). 2-1: the 2 became 1, the 1 went away, and the new pile is 2 (two plates): 2-1 again, nothing changed. 3-2-1: the 3 became 2, the 2 became 1, the 1 went away, and the new pile has 3 counters because there were three plates: 3-2-1 again. Ask: "So which tower is the new one in the picture after the move?" (The tallest one.) "Why is it the tallest?" (Every old tower got shorter, and the new tower is as tall as the number of plates.)

Problem 10 (every way with four; which never appears on the right). Five ways: 4; 3-1; 2-2; 2-1-1; 1-1-1-1. After one move: $4 \rightarrow 3-1$; $3-1 \rightarrow 2-2$; $2-2 \rightarrow 2-1-1$; $2-1-1 \rightarrow 3-1$; $1-1-1-1 \rightarrow 4$. No way stays the same. The only way that never appears on the right is 1-1-1-1; 3-1 appears twice. Hints for finding all the ways, in order: (1) "Start with all four on one plate. Take one off and put it somewhere else. Now what else can you do?" (2) "What is the tallest tower? Can it be 4? 3? 2? 1?" (3) The list by tallest tower: 4 (one way); 3 (only 3-1); 2 (2-2 or 2-1-1); 1 (1-1-1-1). Why 1-1-1-1 never appears after a move: the new pile would be a single counter, so there was only one plate with counters, holding all four, and that plate gives 3-1, not 1-1-1-1. Why nothing with four counters stays still: "we tried all five ways" is a complete answer at this age; the deeper reason (only staircases stay still, and 4 is not 1, 3, 6 or 10) is in section 7.

Problem 11 (staircases). Counts 1, 3, 6, 10. On the big staircase the move takes the top counter off the towers of 4, 3, 2, 1, leaving 3, 2, 1 and nothing, and builds a new tower of 4 on the dotted circles: the towers are 4, 3, 2, 1 again, moved over by one column. Each staircase has one more tower than the one before: 2 more, then 3 more, then 4 more counters. Say: "These are the pictures that stay still: 2-1 with three counters, 3-2-1 with six, 4-3-2-1 with ten." Ask: "How many counters would the next staircase have?" (15: one more tower of 5.)

Problem 12 (races). First race: 4-2 takes 1 move; 3-3 takes 4; 1-1-1-1-1-1 takes 4. Second race: 2-2-1-1 takes 6; 5-1 takes 2; 2-2-2 takes 3. Nothing beats 2-2-1-1: of all eleven ways to place six counters, it is the slowest (six moves), as the table above shows; children will try 4-1-1 (5 moves), 3-1-1-1 (2), 2-1-1-1-1 (3), 6 (3), 3-2-1 (0). When they have tried them all, say so: "You have now tried every one of the eleven ways, and every one ended at 3, 2, 1." The only start that takes exactly one move is 4-2 (3-2-1 itself takes none). Hint for the one-move start: "What must the picture look like one move before 3-2-1? The new pile is the tower of 2, so there were two plates before." Running the races: have all three children make their moves together on your count ("move one . . . move two . . ."), and stop each child when their picture no longer changes; write the number of moves on the scoreboard. The scoreboard's left column is the first race (4-2, 3-3, 1-1-1-1-1-1, top to bottom) and its right column the second race (2-2-1-1, 5-1, 2-2-2); the three empty frames on the next page are for the children's own starts.

Problem 13 (five counters; reserve). $5 \rightarrow 4-1 \rightarrow 3-2 \rightarrow 2-2-1 \rightarrow 3-1-1 \rightarrow 3-2 \rightarrow 2-2-1$: after 2 moves it goes round three pictures (3-2, 2-2-1, 3-1-1) for ever; pictures after 2 and 5 moves are the same, after 3 and 6. $1-1-1-1-1 \rightarrow 5 \rightarrow 4-1 \rightarrow 3-2 \rightarrow 2-2-1$: it joins the first run after one move (at "five on one plate"). Three counters stop; four go round; five go round; six stop.

Problem 14 (ten on one plate; reserve). $10 \rightarrow 9-1 \rightarrow 8-2 \rightarrow 7-2-1 \rightarrow 6-3-1 \rightarrow 5-3-2 \rightarrow 4-3-2-1$; six moves; the last picture is the big staircase and it stays still. One plate of three took 1 move, of six took 3,

of ten took 6: the staircase numbers again.

Problem 15 (which start came before; reserve). $1-1-1 \rightarrow 3$ (the only way). $6 \rightarrow 5-1$ and $2-1-1-1-1 \rightarrow 5-1$ (both). $4-2 \rightarrow 3-2-1$ (the only start other than $3-2-1$ itself). Hint: “The new pile is one of the towers you see, and its height tells how many plates there were before. Which tower is the new one?” For $5-1$: if the new pile is the 1, there was one plate holding all six; if the new pile is the 5, there were five plates: $2-1-1-1-1$.

If the table stalls. A child who does not want to color: you color, the child moves and says the heights. A child who cannot sort the towers by height: let them color in plate order and sort the sizes yourself on the blank line. Restless children: go straight to Problem 12 (the races) and come back. **If the table finishes early:** Problems 13, 14, 15 in that order; then a guessing game: a child builds any start with six counters, everyone guesses how many moves it will take, then run it (use the table of eleven); then seven counters on one plate, with you writing the sizes: $7 \rightarrow 6-1 \rightarrow 5-2 \rightarrow 4-2-1 \rightarrow 3-3-1 \rightarrow 3-2-2 \rightarrow 3-2-1-1 \rightarrow 4-2-1$, which goes round four pictures for ever.

For the share: the K–1 children show the loop of Problem 3 (four counters go round and round) and the last pictures of Problems 5, 6, 7 (six counters always end at 3, 2, 1), and say the counts of the staircases, 1, 3, 6, 10.

4. Grades 2–3 table: order, answers, arguments, hints

Order and timing: Problems 1–8 are the core (about 52 minutes, more than the 42 minutes of table work, so plan as follows). Rough clock, with table work starting at 0:08: Problems 1–2 by 0:19, 3–4 by 0:30, 5 by 0:40, 6 by 0:48. Problems 7 and 8 are the explain step and should not be dropped: if the map (Problem 6) is not finished by 0:44, stop it there, state the law of Problem 7 orally in two minutes using the children's own strips, and spend the last eight minutes on Problem 8. If the table is slow earlier, give each child only two or three of the eleven starts in Problem 5 and pool the results. Problems 9–13 follow in order for a table that is ahead. Children run the moves with counters and write the arrangements; insist that the starting arrangement is rebuilt with counters before each run and that every box is written before the next move.

Watch for: writing an arrangement in the order the piles happen to lie instead of biggest first (then two boxes can look different while the piles are the same); forgetting that a pile of one vanishes; making the new pile from the counters of only some piles.

Problem 1. $6 \rightarrow 5-1 \rightarrow 4-2 \rightarrow 3-2-1 \rightarrow 3-2-1$: 3 moves.

Problem 2. $1-1-1-1-1 \rightarrow 6 \rightarrow 5-1 \rightarrow 4-2 \rightarrow 3-2-1$ (4 moves); $5-1 \rightarrow 4-2 \rightarrow 3-2-1$ (2); $3-3 \rightarrow 2-2-2 \rightarrow 3-1-1-1 \rightarrow 4-2 \rightarrow 3-2-1$ (4); $2-2-1-1 \rightarrow 4-1-1 \rightarrow 3-3 \rightarrow 2-2-2 \rightarrow 3-1-1-1 \rightarrow 4-2 \rightarrow 3-2-1$ (6). All stop at 3-2-1. Most moves: 2-2-1-1; fewest: 5-1. The arrangement in all five runs: 4-2 (5-1 is in three of them, 3-3 in two). Why 4-2 must be in every run: the last move before 3-2-1 must come from an arrangement that becomes 3-2-1, and 4-2 is the only such arrangement other than 3-2-1 itself (Problem 12 proves this). Also worth pointing out: the 3-3 run sits inside the 2-2-1-1 run, and the 6 run inside the all-ones run; once two runs share an arrangement they agree for ever, because the move is the same every time.

Problem 3. $4 \rightarrow 3-1 \rightarrow 2-2 \rightarrow 2-1-1 \rightarrow 3-1 \rightarrow 2-2 \rightarrow 2-1-1 \rightarrow 3-1$. The first arrangement to appear a second time is 3-1: after 1 move and again after 4 moves. Three arrangements go round: 3-1, 2-2, 2-1-1. $1-1-1-1 \rightarrow 4 \rightarrow 3-1 \rightarrow 2-2 \rightarrow 2-1-1$ and $2-2 \rightarrow 2-1-1 \rightarrow 3-1 \rightarrow 2-2$: the same loop. (All five arrangements of four counters end in this loop.)

Problem 4. $5 \rightarrow 4-1 \rightarrow 3-2 \rightarrow 2-2-1 \rightarrow 3-1-1 \rightarrow 3-2$: a loop of three (3-2, 2-2-1, 3-1-1) entered after 2 moves. $2-1-1-1 \rightarrow 4-1 \rightarrow 3-2 \rightarrow 2-2-1 \rightarrow 3-1-1 \rightarrow 3-2$: the same loop, after 2 moves. $3 \rightarrow 2-1 \rightarrow 2-1$: stops after 1 move. Table: 3 R, 4 L, 5 L, 6 R. Ask: "Guess: does 7 stop or loop? Does 10?" (Problems 9 and 8 will tell: 7 loops, 10 stops.)

Problem 5. The eleven ways, listed by biggest pile: 6; 5-1; 4-2, 4-1-1; 3-3, 3-2-1, 3-1-1-1; 2-2-2, 2-2-1-1, 2-1-1-1-1; 1-1-1-1-1-1. Hint for completeness, in order: (1) "What is the biggest pile? Go down from 6 to 1." (2) "With biggest pile 4, the other two counters can be together or apart." (3) "With biggest pile 3, the other three counters: 3, 2-1 or 1-1-1." Moves to reach 3-2-1: 3-2-1: 0; 4-2: 1; 5-1: 2; 3-1-1-1: 2; 6: 3; 2-2-2: 3; 2-1-1-1-1: 3; 1-1-1-1-1-1: 4; 3-3: 4; 4-1-1: 5; 2-2-1-1: 6 (the full runs are in the table of eleven in section 3). New runs to write (the others are on Problems 1–2): $4-2 \rightarrow 3-2-1$; $4-1-1 \rightarrow 3-3 \rightarrow \dots$; $3-2-1 \rightarrow 3-2-1$; $3-1-1-1 \rightarrow 4-2 \rightarrow 3-2-1$; $2-2-2 \rightarrow 3-1-1-1 \rightarrow 4-2 \rightarrow 3-2-1$; $2-1-1-1-1 \rightarrow 5-1 \rightarrow 4-2 \rightarrow 3-2-1$. Every start stops at 3-2-1: yes. Most moves: 2-2-1-1 (6). Exactly two moves: 5-1 and 3-1-1-1. Split the six new starts into two lots of three and give each lot to two children, so every run is done twice and checked; then have everyone copy the other lot's rows so that every child holds the complete table. Say when the table is filled: "This table is a proof. There are only eleven starts, and every one of them stops at 3-2-1. Nobody has to guess any more."

Problem 6. The finished map (bottom to top): level 0: 3-2-1 (with its loop arrow); level 1: 4-2; level 2: 5-1 (left) and 3-1-1-1 (right); level 3: 6 (far left) and 2-1-1-1-1 (left of center), both into 5-1, and 2-2-2 (right)

into 3-1-1-1; level 4: 1-1-1-1-1-1 (far left) into 6, and 3-3 (right) into 2-2-2; level 5: 4-1-1 into 3-3; level 6: 2-2-1-1 into 4-1-1. No arrow coming in: 2-2-1-1, 2-1-1-1-1, 1-1-1-1-1-1 (the top boxes). Two arrows coming in: 4-2 (from 5-1 and 3-1-1-1) and 5-1 (from 6 and 2-1-1-1-1); 3-2-1 as well if the loop arrow is counted (from 4-2 and from itself). Longest path: 6 arrows, from 2-2-1-1. Hints, in order: (1) "Which box must 4-2 be in? It is the only arrangement that becomes 3-2-1." (2) "Which two arrangements become 4-2?" (3) "Fill from the bottom up, one arrow at a time, using the table." Ask afterwards: "Why does the map have no separate pieces?" (Every start's run reaches 3-2-1, so every box is connected to the bottom box.)

Problem 7. In every row the number of piles equals the size of the new pile, because the new pile got exactly one counter from each pile. 4-4-2 (3 piles) $\rightarrow$ new pile 3, giving 3-3-3-1; (4 piles) $\rightarrow$ 4, giving 4-2-2-2; (4 piles) $\rightarrow$ 4, giving 4-3-1-1-1; (5 piles) $\rightarrow$ 5, giving 5-3-2; (3 piles) $\rightarrow$ 3, giving 4-3-2-1. Say the law in one sentence and have the children say it back: "The new pile is as big as the number of piles."

Problem 8. 3 became 2; 2 became 1; the pile of 1 vanished; the new pile has 3 counters (three piles): 3-2-1 again. Ten counters: 4-3-2-1 (check: 4, 3, 2, 1 become 3, 2, 1, nothing, and the new pile is 4). Four counters: 4 $\rightarrow$ 3-1; 3-1 $\rightarrow$ 2-2; 2-2 $\rightarrow$ 2-1-1; 2-1-1 $\rightarrow$ 3-1; 1-1-1-1 $\rightarrow$ 4; none stays. Why there is none: "I tried all five ways" is a complete argument (Problem 3 and this table list them). The argument that also explains 3-2-1 and 4-3-2-1, to build with the children who are ready: if a picture stays the same, the biggest pile after the move must be the new pile, because every old pile got smaller; so the biggest pile equals the number of piles (the law). The old biggest pile lost one counter and is now the second biggest; the old second biggest is now the third; and so on, down to a pile of 1 that vanished and was replaced by the new pile. So the piles are $p, p-1, \dots, 2, 1$: a staircase, and the number of counters is $1 + 2 + \dots + p$. Four is not 1, 3, 6 or 10, so no arrangement of four stays still. (The step "the old biggest is now the second biggest" is where a careful adult fills in the matching argument written out in section 7; third graders can take it as seen on the counters.) Hints, in order: (1) "After the move, is any old pile as big as it was?" (2) "So which pile is the biggest one now?" (3) "How big is the new pile?" (the number of piles) (4) "Where did the old biggest pile go?" (5) "Keep going down."

Problem 9. $7 \rightarrow 6-1 \rightarrow 5-2 \rightarrow 4-2-1 \rightarrow 3-3-1 \rightarrow 3-2-2 \rightarrow 3-2-1-1 \rightarrow 4-2-1$: a loop of four (4-2-1, 3-3-1, 3-2-2, 3-2-1-1), entered after 3 moves. $4-3 \rightarrow 3-2-2 \rightarrow 3-2-1-1 \rightarrow 4-2-1 \rightarrow 3-3-1 \rightarrow 3-2-2$: the same loop, after 1 move. $2-2-2-1 \rightarrow 4-1-1-1 \rightarrow 4-3 \rightarrow 3-2-2 \rightarrow \dots$: the same loop, after 3 moves. (Every one of the fifteen arrangements of seven reaches this loop; the longest way in is from 1-1-1-1-1-1-1, four moves.) Point out: each loop arrangement is 3-2-1 with one more counter somewhere: on the 3 (4-2-1), on the 2 (3-3-1), on the 1 (3-2-2), or as a new pile (3-2-1-1).

Problem 10. $8 \rightarrow 7-1 \rightarrow 6-2 \rightarrow 5-2-1 \rightarrow 4-3-1 \rightarrow 3-3-2 \rightarrow 3-2-2-1 \rightarrow 4-2-1-1 \rightarrow 4-3-1$: a loop of four (4-3-1, 3-3-2, 3-2-2-1, 4-2-1-1), entered after 4 moves. $5-3 \rightarrow 4-2-2 \rightarrow 3-3-1-1 \rightarrow 4-2-2$: a loop of two (4-2-2, 3-3-1-1), entered after 1 move. $4-4 \rightarrow 3-3-2 \rightarrow \dots$: the loop of four, after 1 move. $2-2-2-2 \rightarrow 4-1-1-1-1 \rightarrow 5-3 \rightarrow 4-2-2 \rightarrow 3-3-1-1 \rightarrow 4-2-2$: the loop of two, after 3 moves. Two different loops. This is the first number of counters where different starts end in different places; of the 22 arrangements of eight, 15 reach the loop of four and 7 reach the loop of two (the seven: 4-2-2, 3-3-1-1, 5-3, 6-1-1, 4-1-1-1-1, 2-2-1-1-1-1, 2-2-2-2). Say: "So the answer to 'does every start end in the same place?' is yes for 3, 4, 5, 6 and 7 counters, and no for 8."

Problem 11. Counts 1, 3, 6, 10, 15. Nine: $9 \rightarrow 8-1 \rightarrow 7-2 \rightarrow 6-2-1 \rightarrow 5-3-1 \rightarrow 4-3-2 \rightarrow 3-3-2-1 \rightarrow 4-2-2-1 \rightarrow 4-3-1-1 \rightarrow 4-3-2$, a loop of four entered after 5 moves. Table: R for 1, 3, 6, 10; L for 2, 4, 5, 7, 8, 9. (1: a single counter is a pile of 1 that vanishes and comes back as the new pile: unchanged. $2 \rightarrow 1-1 \rightarrow 2$: a loop of two.) Any try for 7, 8 or 9 changes, for example $4-2-1 \rightarrow 3-3-1$, $4-3-1 \rightarrow 3-3-2$, $4-3-2 \rightarrow 3-3-2-1$; the reason is Problem 8's argument: only staircases stay still, and 7, 8, 9 are not staircase numbers. Ask: "Which numbers between 10 and 30 will get an R?" (15, 21, 28.)

Problem 12. Parents of 4-2: 5-1 and 3-1-1-1. Parents of 5-1: 6 and 2-1-1-1-1. Parents of 4-3-2-1: 5-3-2 and 4-3-2-1 itself. Why 2-2-1-1 has no parent: the new pile is one of the piles, so it has at most 2 counters, so before the move there were at most 2 piles; a move from at most 2 piles leaves at most 3 piles (the old ones, minus any that vanished, plus the new one); 2-2-1-1 has four piles. The general method for parents, if a child wants it: choose which pile is the new one; its size m is the number of piles before the move; take that pile away, give one counter back to every other pile, and add single counters until there are m piles; if there are already more than m other piles, that choice is impossible. For 4-3-2-1: new pile 4 gives 4-3-2 plus one single, 4-3-2-1 itself; new pile 3 gives 5-3-2; new pile 2 or 1 is impossible (three other piles remain, more than 2 or 1).

Problem 13. One pile of 3: 1 move; of 6: 3 moves; of 10: 6 moves ($10 \rightarrow 9-1 \rightarrow 8-2 \rightarrow 7-2-1 \rightarrow 6-3-1 \rightarrow 5-3-2 \rightarrow 4-3-2-1$). The numbers 1, 3, 6 are the staircase numbers, so the prediction is 10, and it is right: $15 \rightarrow 14-1 \rightarrow 13-2 \rightarrow 12-2-1 \rightarrow 11-3-1 \rightarrow 10-3-2 \rightarrow 9-3-2-1 \rightarrow 8-4-2-1 \rightarrow 7-4-3-1 \rightarrow 6-4-3-2 \rightarrow 5-4-3-2-1$, ten moves. Point out on the run: the piles other than the biggest one form 1, then 2-1, then 3-2-1, then 4-3-2-1 after 1, 3, 6 and 10 moves; the reason is in section 7.

If the table stalls. On Problem 5, give each child two starts and let the table pool the results. On Problem 6, fill the bottom three boxes together, then let them continue. On Problem 8, accept “we tried all five” and move on; come back to the argument at the share if there is time. **If the table finishes early:** after Problem 13, run every two-pile start with ten counters: 9-1 takes 5 moves, 8-2 takes 4, 7-3 takes 8, 6-4 takes 2, 5-5 takes 6 (the most equal start is not the fastest); then look for a start with ten counters that takes more than ten moves: 6-2-1-1, 5-2-2-1 and 5-2-1-1-1 take 11; 3-3-2-1-1, 3-2-2-2-1 and 3-2-2-1-1-1 take 12, and 12 is the most any start of ten takes (checked over all 42 starts). Example: $3-3-2-1-1 \rightarrow 5-2-2-1 \rightarrow 4-4-1-1 \rightarrow 4-3-3 \rightarrow 3-3-2-2 \rightarrow 4-2-2-1-1 \rightarrow 5-3-1-1 \rightarrow 4-4-2 \rightarrow 3-3-3-1 \rightarrow 4-2-2-2 \rightarrow 4-3-1-1-1 \rightarrow 5-3-2 \rightarrow 4-3-2-1$.

For the share: this table shows the map of six (every start flows to 3-2-1) and, if reached, the two loops of eight.

5. Grades 4–5 table: order, answers, arguments, hints

Order and timing: Problems 1–6 are the core (about 49 minutes; the three children share the fifteen starts of Problem 5, five each); then 7–13 in order. Rough clock, with table work starting at 0:08: Problems 1–2 by 0:24, 3–4 by 0:36, 5 by 0:46, 6 by 0:50 at the latest. If Problem 5 is running long, skip drawing the map of seven: its three questions can be answered from the strips, and the point of Problem 6 (two loops for eight) must be reached. Problems 4, 8, 9 and 12 are where the mathematician pushes for an argument; the others are doing and recording.

Problem 1. $6 \rightarrow 5-1 \rightarrow 4-2 \rightarrow 3-2-1$ (3 moves); $3-3 \rightarrow 2-2-2 \rightarrow 3-1-1-1 \rightarrow 4-2 \rightarrow 3-2-1$ (4); $4-1-1 \rightarrow 3-3 \rightarrow 2-2-2 \rightarrow 3-1-1-1 \rightarrow 4-2 \rightarrow 3-2-1$ (5); $2-2-1-1 \rightarrow 4-1-1 \rightarrow 3-3 \rightarrow \dots \rightarrow 3-2-1$ (6). All stop at $3-2-1$ through $4-2$; the $4-1-1$ run contains the $3-3$ run, and the $2-2-1-1$ run contains the $4-1-1$ run. Table for $2-2-1-1$: $2-2-1-1$ (4 piles) $\rightarrow$ new pile 4; $4-1-1$ (3) $\rightarrow$ 3; $3-3$ (2) $\rightarrow$ 2; $2-2-2$ (3) $\rightarrow$ 3; $3-1-1-1$ (4) $\rightarrow$ 4; $4-2$ (2) $\rightarrow$ 2. The number of piles equals the size of the new pile, always: the new pile has exactly one counter from each pile. Have the children say this law aloud; it is used in Problems 4, 8, 9, 12.

Problem 2. The eleven ways by biggest pile: 6; 5-1; 4-2, 4-1-1; 3-3, 3-2-1, 3-1-1-1; 2-2-2, 2-2-1-1, 2-1-1-1-1; 1-1-1-1-1-1 ($1 + 1 + 2 + 3 + 3 + 1 = 11$). Moves: $6 \rightarrow 5-1$; $5-1 \rightarrow 4-2$; $4-2 \rightarrow 3-2-1$; $4-1-1 \rightarrow 3-3$; $3-3 \rightarrow 2-2-2$; $3-2-1 \rightarrow 3-2-1$; $3-1-1-1 \rightarrow 4-2$; $2-2-2 \rightarrow 3-1-1-1$; $2-2-1-1 \rightarrow 4-1-1$; $2-1-1-1-1 \rightarrow 5-1$; $1-1-1-1-1-1 \rightarrow 6$. The map is one tree: above $3-2-1$ sits $4-2$; above $4-2$ sit $5-1$ and $3-1-1-1$; above $5-1$ sit 6 and $2-1-1-1-1$; above 6 sits $1-1-1-1-1-1$; above $3-1-1-1$ the chain $2-2-2$, $3-3$, $4-1-1$, $2-2-1-1$. Every path leads to $3-2-1$. Say: “That is a complete proof that every arrangement of six counters ends at $3-2-1$: there are only eleven, and here they all are.” Hints: (1) “Who becomes $3-2-1$? Only $4-2$. Put it first.” (2) “Now who becomes $4-2$?”

Problem 3. Moves to $3-2-1$: $3-2-1$: 0; $4-2$: 1; $5-1$: 2; $3-1-1-1$: 2; 6 : 3; $2-2-2$: 3; $2-1-1-1-1$: 3; $1-1-1-1-1-1$: 4; $3-3$: 4; $4-1-1$: 5; $2-2-1-1$: 6. Most: $2-2-1-1$. Exactly three moves: three arrangements (6 , $2-2-2$, $2-1-1-1-1$). No arrow in: $2-2-1-1$, $2-1-1-1-1$, $1-1-1-1-1-1$. Two arrows in: $4-2$, $5-1$, and $3-2-1$ (from $4-2$ and its own loop).

Problem 4. The move on $3-2-1$: $3 \rightarrow 2$, $2 \rightarrow 1$, 1 vanishes, new pile 3 (three piles). The argument, and the answers expected: no old pile is as big as it was (each lost a counter, and the piles of 1 are gone). So the biggest pile of the new arrangement cannot be an old pile; it must be the new pile. The new pile’s size is the number of piles (the law), so in an arrangement that does not change, the biggest pile equals the number of piles, call it p . Next: the new arrangement has the new pile p plus every old pile reduced by one, the old piles of 1 having vanished; since the number of piles is still p and one pile was added, exactly one old pile vanished, so there was exactly one pile of 1. Set the new pile p aside from the new arrangement and the old biggest pile p aside from the old arrangement; what is left must match: on the new side, every old pile except the 1, each one counter smaller; on the old side, every old pile except the biggest. So the second biggest old pile is the biggest minus one, the third is the second minus one, and so on, down to the smallest, which is the 1. The arrangement is $p, p-1, \dots, 2, 1$, a staircase, with $1 + 2 + \dots + p$ counters. Numbers from 1 to 30: 1, 3, 6, 10, 15, 21, 28. Hints, in order: (1) “Can the biggest pile after the move be one of the old piles?” (2) “How big is the new pile?” (3) “Where is the old biggest pile now? Is anything else that big?” (4) “And the old second biggest?” (5) “What happens at the bottom?” The children need only produce the chain “biggest = number of piles; then one less, one less, $\dots$, down to 1”; the mathematician supplies the matching step if asked why the second biggest cannot be anything else. The written proof is in section 7.

Problem 5. The fifteen ways by biggest pile: 7; 6-1; 5-2, 5-1-1; 4-3, 4-2-1, 4-1-1-1; 3-3-1, 3-2-2, 3-2-1-1, 3-1-1-1-1; 2-2-2-1, 2-2-1-1-1, 2-1-1-1-1-1; 1-1-1-1-1-1-1. One loop of four: $4-2-1 \rightarrow 3-3-1 \rightarrow 3-2-2 \rightarrow$

3-2-1-1 $\rightarrow$ 4-2-1. Runs: 7 $\rightarrow$ 6-1 $\rightarrow$ 5-2 $\rightarrow$ 4-2-1 (3 moves in); 1-1-1-1-1-1 $\rightarrow$ 7 (4 moves, the farthest); 2-1-1-1-1-1 $\rightarrow$ 6-1 (3); 3-1-1-1-1 $\rightarrow$ 5-2 (2); 5-1-1 $\rightarrow$ 4-3 $\rightarrow$ 3-2-2 (2); 2-2-1-1-1 $\rightarrow$ 5-1-1 (3); 4-1-1-1 $\rightarrow$ 4-3 (2); 2-2-2-1 $\rightarrow$ 4-1-1-1 (3); 4-3 $\rightarrow$ 3-2-2 (1); 5-2 $\rightarrow$ 4-2-1 (1); 6-1 $\rightarrow$ 5-2 (2). Every arrangement of seven reaches the loop. No arrow in: 3-1-1-1-1, 2-2-2-1, 2-2-1-1-1, 2-1-1-1-1-1, 1-1-1-1-1-1-1. The map is a tree hanging on a ring of four instead of on a point. Hint for the map: “Draw the loop first, as a ring, then hang each run on the place where it enters.”

Problem 6. 8 $\rightarrow$ 7-1 $\rightarrow$ 6-2 $\rightarrow$ 5-2-1 $\rightarrow$ 4-3-1 $\rightarrow$ 3-3-2 $\rightarrow$ 3-2-2-1 $\rightarrow$ 4-2-1-1 $\rightarrow$ 4-3-1: loop A of four (4-3-1, 3-3-2, 3-2-2-1, 4-2-1-1), entered after 4 moves. 5-3 $\rightarrow$ 4-2-2 $\rightarrow$ 3-3-1-1 $\rightarrow$ 4-2-2: loop B of two, after 1 move. 4-4 $\rightarrow$ 3-3-2: loop A after 1 move. 2-2-2-2 $\rightarrow$ 4-1-1-1-1 $\rightarrow$ 5-3 $\rightarrow$ 4-2-2: loop B after 3 moves. 6-1-1 $\rightarrow$ 5-3 $\rightarrow$ 4-2-2: loop B after 2 moves. Two different loops, of lengths 4 and 2. This breaks the pattern of 3 to 7 counters: where you end depends on where you start.

Problem 7. The 22 ways by biggest pile: 8; 7-1; 6-2, 6-1-1; 5-3, 5-2-1, 5-1-1-1; 4-4, 4-3-1, 4-2-2, 4-2-1-1, 4-1-1-1-1; 3-3-2, 3-3-1-1, 3-2-2-1, 3-2-1-1-1, 3-1-1-1-1-1; 2-2-2-2, 2-2-2-1-1, 2-2-1-1-1-1, 2-1-1-1-1-1-1; 1-1-1-1-1-1-1-1. Loop B (of two) is reached by seven starts: 4-2-2 and 3-3-1-1 (0 moves), 5-3 (1), 6-1-1 (2), 4-1-1-1-1 (2), 2-2-2-2 (3), 2-2-1-1-1-1 (3). Loop A (of four) is reached by the other fifteen: 4-3-1, 3-3-2, 3-2-2-1, 4-2-1-1 (0), 5-2-1 (1), 4-4 (1), 6-2 (2), 5-1-1-1 (2), 3-2-1-1-1 (2), 7-1 (3), 3-1-1-1-1-1 (3), 2-2-2-1-1 (3), 8 (4), 2-1-1-1-1-1-1 (4), 1-1-1-1-1-1-1-1 (5). Runs not yet on the strips: 5-1-1-1 $\rightarrow$ 4-4; 3-2-1-1-1 $\rightarrow$ 5-2-1; 2-2-2-1-1 $\rightarrow$ 5-1-1-1; 2-2-1-1-1-1 $\rightarrow$ 6-1-1; 3-1-1-1-1-1 $\rightarrow$ 6-2; 2-1-1-1-1-1-1 $\rightarrow$ 7-1; 1-1-1-1-1-1-1-1 $\rightarrow$ 8; 7-1 $\rightarrow$ 6-2; 4-1-1-1-1 $\rightarrow$ 5-3.

Problem 8. Parents of 5-3-2: 6-4, 6-3-1, 4-3-1-1-1. Parents of 4-4: 5-1-1-1 only. Parents of 2-2-2-2: none. The method: the new pile is one of the piles of the child arrangement; choose which. Its size m is the number of piles before the move. Remove that pile, give one counter back to each remaining pile, and add single counters until there are m piles. If more than m piles remain after removing the new pile, that choice is impossible. For 5-3-2: new pile 5 gives 4-3 plus three singles, 4-3-1-1-1; new pile 3 gives 6-3 with three piles needed, so 6-3-1; new pile 2 gives 6-4. For 4-4: new pile 4 gives 5 plus three singles, 5-1-1-1 (both fours give the same). For 2-2-2-2: new pile 2 leaves three piles, more than 2, impossible. Check every answer by building it and making the move. Hints: (1) “The new pile is one of the piles you see. Try each one.” (2) “If the new pile is the 3, how many piles were there before?” (3) “Undo the move: give every other pile its counter back.”

Problem 9. No parent, six counters: 2-2-1-1 (biggest 2, 4 piles), 2-1-1-1-1 (2, 5), 1-1-1-1-1-1 (1, 6). Seven: 3-1-1-1-1 (3, 5), 2-2-2-1 (2, 4), 2-2-1-1-1 (2, 5), 2-1-1-1-1-1 (2, 6), 1-1-1-1-1-1-1 (1, 7). Rule: an arrangement has no parent exactly when its biggest pile is at most the number of piles minus 2 (equivalently: it has a parent exactly when some pile is at least the number of piles minus 1). Why: a move from m piles makes a pile of size m and leaves at most m other piles, so the result has at most $m + 1$ piles and contains a pile of size m : some pile is at least (number of piles) $- 1$. Conversely, if a pile of size m is at least (number of piles) $- 1$, the method of Problem 8 builds a parent. Eight counters, no parent (seven of them): 3-2-1-1-1, 3-1-1-1-1-1, 2-2-2-2, 2-2-2-1-1, 2-2-1-1-1-1, 2-1-1-1-1-1-1, 1-1-1-1-1-1-1-1. Checks: for 3-2-1-1-1, new pile 3 leaves four piles, too many; new pile 2 or 1 leaves four; so no parent. For 2-2-2-2, as in Problem 8. Counts of parentless arrangements for 3, 4, $\dots$, 10 counters: 1, 1, 2, 3, 5, 7, 10, 14.

Problem 10. 10 $\rightarrow$ 9-1 $\rightarrow$ 8-2 $\rightarrow$ 7-2-1 $\rightarrow$ 6-3-1 $\rightarrow$ 5-3-2 $\rightarrow$ 4-3-2-1: 6 moves. 5-5 $\rightarrow$ 4-4-2 $\rightarrow$ 3-3-3-1 $\rightarrow$ 4-2-2-2 $\rightarrow$ 4-3-1-1-1 $\rightarrow$ 5-3-2 $\rightarrow$ 4-3-2-1: 6. 7-3 $\rightarrow$ 6-2-2 $\rightarrow$ 5-3-1-1 $\rightarrow$ 4-4-2 $\rightarrow \dots$: 8. 4-4-1-1 $\rightarrow$ 4-3-3 $\rightarrow$ 3-3-2-2 $\rightarrow$ 4-2-2-1-1 $\rightarrow$ 5-3-1-1 $\rightarrow$ 4-4-2 $\rightarrow \dots$: 10. Twelve moves: 3-3-2-1-1, 3-2-2-2-1 and 3-2-2-1-1-1, and no start of ten counters takes more (checked over all 42). Example: 3-3-2-1-1 $\rightarrow$ 5-2-2-1 $\rightarrow$ 4-4-1-1 $\rightarrow$ 4-3-3 $\rightarrow$ 3-3-2-2 $\rightarrow$ 4-2-2-1-1 $\rightarrow$ 5-3-1-1 $\rightarrow$ 4-4-2 $\rightarrow$ 3-3-3-1 $\rightarrow$ 4-2-2-2 $\rightarrow$ 4-3-1-1-1 $\rightarrow$ 5-3-2 $\rightarrow$ 4-3-2-1. Hints for the search: (1) “A slow start must feed into a slow run: what comes before

4-4-1-1?" (its only parent is 5-2-2-1, which takes 11 moves) (2) "And before that?" (the only parent of 5-2-2-1 is 3-3-2-1-1, which takes 12 and has no parent itself). Eleven-move starts: 6-2-1-1, 5-2-2-1, 5-2-1-1-1; the three twelve-move starts are their parents (3-2-2-1-1-1 $\rightarrow$ 6-2-1-1, 3-3-2-1-1 $\rightarrow$ 5-2-2-1, 3-2-2-2-1 $\rightarrow$ 5-2-1-1-1) and have no parents themselves. The slowest starts for 3, 6, 10 and 15 counters take 2, 6, 12 and 20 moves.

Problem 11. $3 \rightarrow 2-1 \rightarrow 2-1$: 1 move. Prediction: 10 (the pattern 1, 3, 6 is the staircase numbers). Run: $15 \rightarrow 14-1 \rightarrow 13-2 \rightarrow 12-2-1 \rightarrow 11-3-1 \rightarrow 10-3-2 \rightarrow 9-3-2-1 \rightarrow 8-4-2-1 \rightarrow 7-4-3-1 \rightarrow 6-4-3-2 \rightarrow 5-4-3-2-1$: 10 moves. The small piles form 1, 2-1, 3-2-1, 4-3-2-1 after 1, 3, 6, 10 moves: the small piles play their own game in which the new pile gets one extra counter (from the big pile), and that game builds one more diagonal of the staircase every time; meanwhile the big pile loses exactly one counter per move. Fifteen ones: 11 moves, because the first move makes one pile of fifteen and then the run above follows. The proof is in section 7.

Problem 12. $9 \rightarrow 8-1 \rightarrow 7-2 \rightarrow 6-2-1 \rightarrow 5-3-1 \rightarrow 4-3-2 \rightarrow 3-3-2-1 \rightarrow 4-2-2-1 \rightarrow 4-3-1-1 \rightarrow 4-3-2$: a loop of four entered after 5 moves. The loops as staircase plus extras (an extra written +1 on the step it sits on, and a fourth position for "a new pile of 1"):

- Seven: $4-2-1 = (3+1)-2-1$; $3-3-1 = 3-(2+1)-1$; $3-2-2 = 3-2-(1+1)$; $3-2-1-1 = 3-2-1-(0+1)$. One extra, which moves one step to the right each move and, from the last position, comes back to the first: four positions, so a loop of four.
- Eight, loop A: $4-3-1 = (3+1)-(2+1)-1$; $3-3-2 = 3-(2+1)-(1+1)$; $3-2-2-1 = 3-2-(1+1)-(0+1)$; $4-2-1-1 = (3+1)-2-1-(0+1)$. Two extras side by side, moving together round four positions: a loop of four. Loop B: $4-2-2 = (3+1)-2-(1+1)$; $3-3-1-1 = 3-(2+1)-1-(0+1)$. Two extras opposite each other: after two moves the pattern is back, a loop of two.
- Nine: $4-3-2 = (3+1)-(2+1)-(1+1)$; $3-3-2-1 = 3-(2+1)-(1+1)-(0+1)$; $4-2-2-1 = (3+1)-2-(1+1)-(0+1)$; $4-3-1-1 = (3+1)-(2+1)-1-(0+1)$. Three extras, one empty position going round: a loop of four.

It works for every loop arrangement. Why the extras move one step to the right: the new pile becomes the new first step (it has 3 counters, plus 1 if there was a pile of 1 made of an extra); the old first step (3 or 4) becomes the second step (2 or 3), keeping its extra; the old second becomes the third; the old third step (1 or 2) becomes nothing or a pile of 1, keeping its extra. So each extra shifts one position, and the extra at the last position reappears at the first. The loop length is the number of moves until the pattern of extras comes back: a single extra or a single gap needs four moves; two adjacent extras need four; two opposite extras need two. Prediction for twelve counters (4-3-2-1 with two extras on five positions): two loops, both of length five, one with the extras side by side and one with the extras one apart; runs: $12 \rightarrow 11-1 \rightarrow 10-2 \rightarrow 9-2-1 \rightarrow 8-3-1 \rightarrow 7-3-2 \rightarrow 6-3-2-1 \rightarrow 5-4-2-1$ (the loop 5-4-2-1, 4-4-3-1, 4-3-3-2, 4-3-2-2-1, 5-3-2-1-1), and $4-4-4 \rightarrow 3-3-3-3 \rightarrow 4-2-2-2-2 \rightarrow 5-3-1-1-1-1 \rightarrow 6-4-2 \rightarrow 5-3-3-1$ (the loop 5-3-3-1, 4-4-2-2, 4-3-3-1-1, 5-3-2-2, 4-4-2-1-1). Offer this as the next challenge if the table gets here.

Problem 13. Start: left A-2-3 (top to bottom), middle 4-5, right 6. After 1 move: left A-4-6, middle 2-3, right 5. After 2: left A-2-5, middle 4-6, right 3. After 3: left A-4-3, middle 2-5, right 6. After 4: left A-2-6, middle 4-3, right 5. After 5: left A-4-5, middle 2-6, right 3. After 6: left A-2-3, middle 4-5, right 6, the start. The ace never moves. Back after 2 moves: 2 and 4 (and the ace). Back after 3: 3, 5, 6. Every card back after 6 moves. A card's period is (its pile's position from the left) + (its depth in the pile) - 1: the ace 1; the 2 and the 4 have period 2; the 3, 5, 6 have period 3. Ten cards: after 1 move the piles are A-5-8-10, 2-3-4, 6-7, 9; the ace is fixed; 2 and 5 have period 2; 3, 6, 8 period 3; 4, 7, 9, 10 period 4; all ten back after 12 moves (the least common multiple of 1, 2, 3, 4). With fifteen cards it would be 60 moves.

If the table stalls. On Problem 2, place 4-2 and its two parents for them. On Problem 4, accept the chain

“biggest pile = number of piles, then one less each time” and let the mathematician fill the gap about the vanishing pile of 1. On Problem 5, if listing fifteen is slow, give the list and let them run and map. **If the table finishes early:** after Problem 13, the twelve-counter prediction above; then the backward tree of ten: start at 4-3-2-1, find its parent 5-3-2, then the parents of 5-3-2 (6-4, 6-3-1, 4-3-1-1-1), then their parents ($6-4 \leftarrow 7-1-1-1$, $5-1-1-1-1$; $6-3-1 \leftarrow 7-2-1$, $4-2-1-1-1$; $4-3-1-1-1 \leftarrow 4-2-2-2$), and so on, as far as it goes: all 42 arrangements of ten are reached, the longest chains have 12 arrows and end at 3-3-2-1-1, 3-2-2-2-1 and 3-2-2-1-1-1.

For the share: this table explains, with counters, why only staircases stay still (Problem 4), and shows the cards of Problem 13 if it got there.

6. The share (eight minutes) and closing

Each table shows one thing, two minutes each: K–1 shows the four-counter loop and the last pictures of the six-counter runs, and says the staircase counts 1, 3, 6, 10; grades 2–3 hold up the map of six and, if they have it, the two loops of eight; grades 4–5 give the argument that a picture that does not change must be a staircase.

The organizer closes in one minute: “Three, six and ten counters always stop, and they stop at a staircase: 2-1, 3-2-1, 4-3-2-1. Four, five, seven, eight and nine never stop; they go round loops, and with eight counters there are two different loops. The numbers that stop are the staircase numbers 1, 3, 6, 10, 15, 21, and so on. That every start with a staircase number of counters ends at the staircase is a theorem proved in 1982 by Jørgen Brandt; you proved it today for 3 and 6 by trying every start.”

7. The mathematics behind the session (for the two mathematicians)

Setting. An arrangement of n counters is a partition of n . The move (Bulgarian solitaire) is a map on the finite set of partitions of n , so every orbit is eventually periodic: it ends at a fixed point or in a cycle. The one law every argument uses: the new pile has as many counters as there were piles.

Fixed points are exactly the staircases (K–1 Problem 9, grades 2–3 Problem 8, grades 4–5 Problem 4). Let $\lambda = (\lambda_1 \geq \dots \geq \lambda_p)$ be unchanged by the move. After the move the piles are $\lambda_i - 1$ for each $\lambda_i \geq 2$, plus the new pile p . Every old pile shrank, so the largest pile afterwards, which is again λ_1 , must be the new pile: $\lambda_1 = p$. The number of piles afterwards is (number of $\lambda_i \geq 2$) + 1 and must be p , so exactly one λ_i equals 1: $\lambda_p = 1$ and $\lambda_{p-1} \geq 2$. Remove one pile of size p from the old arrangement and the new pile p from the new one; what remains must agree as multisets: $\{\lambda_1 - 1, \dots, \lambda_{p-1} - 1\} = \{\lambda_2, \dots, \lambda_p\}$. Sorted, $\lambda_1 - 1 = \lambda_2$, $\lambda_2 - 1 = \lambda_3$, $\dots$, $\lambda_{p-1} - 1 = \lambda_p = 1$. Hence $\lambda = (p, p-1, \dots, 1)$. Conversely every staircase is fixed. So a fixed point exists exactly when n is triangular, and then it is unique. Verified by brute force for $n \leq 30$.

Parents and Garden of Eden states (grades 2–3 Problem 12, grades 4–5 Problems 8 and 9). μ is the image of some λ if and only if μ has a part $m \geq \ell(\mu) - 1$, where ℓ is the number of parts; the parents are obtained, for each such distinct part size m , by removing one part m , adding 1 to every other part, and adding $m - (\ell(\mu) - 1)$ parts equal to 1. Proof: if λ has m parts, the move creates a part m and leaves at most m other parts, so $\ell(\mu) \leq m + 1$; conversely the construction gives a λ with exactly m parts whose image is μ , and every parent arises this way, since the new part is one of the parts of μ and its size fixes $\ell(\lambda)$. Consequently μ has no parent exactly when its largest part is at most $\ell(\mu) - 2$. Counts of parentless partitions for $n = 3, \dots, 12$: 1, 1, 2, 3, 5, 7, 10, 14, 20, 27; they agree with $p(n-3)$ up to $n = 8$ and fall behind from $n = 9$ (10 against 11). Both rules verified against brute force for $n \leq 18$.

Brandt's theorem (cited; verified by computer for $n \leq 20$). J. Brandt, "Cycles of partitions", Proc. Amer. Math. Soc. 85 (1982): if $n = T_k = k(k+1)/2$, every start reaches the staircase $(k, \dots, 1)$; if $n = T_k + r$ with $0 < r \leq k$, every start ends in a cycle, and the cycle states are exactly the $\binom{k+1}{r}$ partitions obtained from $(k, k-1, \dots, 1, 0)$ by adding 1 to r of its $k+1$ entries. The part provable at the table (grades 4–5 Problem 12): write such a state as $(k + \varepsilon_0, k - 1 + \varepsilon_1, \dots, 1 + \varepsilon_{k-1}, \varepsilon_k)$ with $\varepsilon \in \{0, 1\}^{k+1}$, $\sum \varepsilon = r$, the piles listed newest first (this order is also weakly decreasing, so it is the order the children write). It has $k + \varepsilon_k$ piles. The move sends it to $(k + \varepsilon_k, k - 1 + \varepsilon_0, \dots, 1 + \varepsilon_{k-2}, \varepsilon_{k-1})$: the same form with ε cyclically shifted one place. So this family is closed under the move, the move acts on it as a rotation of the necklace ε , cycle lengths divide $k+1$, and the number of cycles is the number of binary necklaces of length $k+1$ with r ones. There is a single cycle exactly when $r \in \{0, 1, k\}$: $n = 2, 4, 5, 7, 9, 11, 14, 16, 20$ have one loop, and $n = 8, 12, 13, 17, 18, 19$ have several (8: necklaces 1100 and 1010, loops of 4 and 2; 12 and 13: two loops of 5; 17: loops of 6, 6, 3; 18: loops of 6, 6, 6, 2). The hard half of the theorem, that every start reaches this family, is what the exhaustive maps establish for $n \leq 8$ and $n = 10$ in the session; the general proof is in Brandt's paper and in E. Akin and M. Davis, "Bulgarian solitaire", Amer. Math. Monthly 92 (1985).

Young diagrams and anti-diagonals (grades 4–5 Problem 13). Draw a partition as rows of cells, newest pile on top, the first counter of each row at the left. The move removes the first cell of every row and lays those cells down as a new top row, the cell from row i in column i . A cell in row i , column j goes to row $i+1$, column $j-1$ if $j \geq 2$, and to row 1, column i if $j = 1$; either way $i+j$ is unchanged, so every cell stays on its anti-diagonal. At the staircase the first k anti-diagonals are full and the move rotates the d cells of anti-diagonal d cyclically; a card in pile i (from the left) at depth j has period $i+j-1$, and all cards return after $\text{lcm}(1, \dots, k)$ moves: 6 for $k = 3$, 12 for $k = 4$, 60 for $k = 5$. The extras of a cycle state are cells on anti-diagonal $k+1$, and the necklace rotation above is the rotation of that diagonal.

How long it takes. For $n = T_k$ the number of moves from any start is at most $k^2 - k$ (K. Igusa, “Solution of the Bulgarian solitaire conjecture”, Math. Mag. 58 (1985); G. Etienne, “Tableaux de Young et solitaire bulgare”, J. Combin. Theory Ser. A 57 (1991)). The bound is attained for $k = 2, \dots, 6$: slowest starts 1-1-1 (2 moves), 2-2-1-1 (6), 3-3-2-1-1 and two others (12), sixteen starts with 15 counters (20), and 30 moves for 21 counters. The distribution for 10 counters, moves: number of starts, is 0:1, 1:1, 2:3, 3:5, 4:5, 5:3, 6:4, 7:4, 8:4, 9:3, 10:3, 11:3, 12:3.

The single pile (grades 2–3 Problem 13, grades 4–5 Problem 11). From one pile of N counters, after m moves the state is $\{N - m\} \cup \tau_m$, where τ_0 is empty and $\tau_{m+1} = \{x - 1 : x \in \tau_m, x \geq 2\} \cup \{|\tau_m| + 1\}$, as long as $N - m$ stays the largest pile: the big pile loses one counter, the small piles lose one each, and the new pile has one counter from the big pile plus one from each small pile. In this “tail game”, starting from the j -staircase $(j, \dots, 1)$, the next $j + 1$ moves give $(j + 1, j - 1, \dots, 1)$, $(j + 1, j, j - 2, \dots, 1)$, $\dots$, $(j + 1, j, \dots, 2)$, $(j + 1, j, \dots, 1)$: the $(j + 1)$ -th diagonal fills one cell per move. So τ reaches the $(k - 1)$ -staircase after $1 + 2 + \dots + (k - 1) = T_{k-1}$ moves, and at that moment the state is $\{T_k - T_{k-1}\} \cup (k - 1, \dots, 1) = (k, k - 1, \dots, 1)$. It cannot have rested earlier, because before that the big pile exceeds k . Hence one pile of T_k takes exactly T_{k-1} moves: 1, 3, 6, 10, 15 for $k = 2, \dots, 6$, and all-ones takes one more.

Where it leads. Partitions and Young diagrams; the conjugate description (in the conjugate partition the move deletes the largest part m and adds 1 to the next m parts, the “shift” map of J. R. Griggs and C.-C. Ho, “The cycling of partitions and compositions under repeated shifts”, Adv. in Appl. Math. 21 (1998)); enumeration of Garden of Eden partitions (B. Hopkins and J. A. Sellers, “Exact enumeration of Garden of Eden partitions”, Integers 7 (2007)); the survey B. Hopkins, “30 years of Bulgarian solitaire”, College Math. J. 43 (2012); variants (Austrian solitaire in Akin and Davis; Montreal solitaire, J. Cannings and J. Haigh, J. Combin. Theory Ser. A 60 (1992)); random Bulgarian solitaire (S. Popov, Random Structures Algorithms 27 (2005)); the popular account is Martin Gardner’s column in Scientific American, August 1983. Undergraduate topics touched: finite dynamical systems (transients, cycles, basins), necklaces and Burnside counting, integer partitions and generating functions, orbits of permutations and least common multiples. The bibliographic details are quoted from memory and were not checked against the originals; every mathematical statement above was checked by computation.